

# Paul Jerabek

Group Leader Computational Materials Design  
Deputy Head of Department of Materials Design

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[Scholar profile](#)

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Helmholtz-Zentrum Hereon, Max-Planck-Straße 1, 21502 Geesthacht

## Education

- Dec 2010 – Sep 2014 **Dr. rer. nat.**, *Philipps-Universität Marburg*  
Thesis: *Clusterartige und hochkoordinierte Metallkomplexe*
- Jun 2010 – Nov 2010 **Dipl.-Chem.**, *Philipps-Universität Marburg*  
Thesis: *Bindungstheoretische Untersuchungen der Systeme des Typs YE-X-EY*
- Oct 2005 – Jun 2010 **Studies of Chemistry**, *Philipps-Universität Marburg*

## Professional experience

- Sep 2022 – present **Deputy Head of Department of Materials Design**, *Helmholtz-Zentrum Hereon*  
Department administration and representation, third-party funding acquisition, research strategy planning, digital transformation.
- Oct 2021 – present **Co-Leader, German-New Zealand Green Hydrogen Centre (He Honoka Hauwai)**, *Helmholtz-Zentrum Hereon / University of Otago*  
Administration and representation, strategic planning, networking, funding acquisition, event organization.
- Sep 2019 – present **Group Leader “Metal Hydride Modeling”**, *Helmholtz-Zentrum Hereon*  
First-principles molecular dynamics, CALPHAD and phase-field simulations for the characterization and optimization of novel metal hydride materials for large-scale hydrogen storage. Close collaborations with Lawrence Livermore National Laboratory (USA) and the University of Otago (New Zealand).
- Jul 2018 – Aug 2019 **Humboldt Return Fellow**, *Max-Planck-Institut für Kohlenforschung*  
Theoretical investigation of physical properties of organesson; substituent and solvent effects on switching temperature in Fe(II) spin-crossover complexes; reaction mechanism exploration for asymmetrical, homogeneous catalysis; bonding situation and electronic properties of pincer complexes.
- Dec 2017 – Jun 2018 **Postdoctoral Research Assistant**, *Massey University*  
Monte-Carlo melting simulations of heavy and superheavy elements; first-principles molecular dynamics simulations of coinage metals in liquid and solid phase; highly accurate calculations of atomic polarizabilities.
- Dec 2015 – Nov 2017 **Feodor Lynen Research Fellow**, *Massey University*  
Photochemistry of small molecules in different environments; influence of relativistic and solvent effects on atomic and molecular bonding properties; development of highly accurate ab-initio many-body potentials for heavy and superheavy elements.
- Dec 2014 – Nov 2015 **Postdoctoral Research Assistant**, *Philipps-Universität Marburg*  
Method development (C) for core electron spectroscopy of organic molecules on surfaces; density functional theory based investigation of transition metal clusters and complexes.

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| Dec 2010 – Nov 2014 | <b>Research Assistant, <i>Philipps-Universität Marburg</i></b><br>Quantum chemical investigation of main-group and transition metal compounds in collaboration with experimental groups in Bochum, München, Göttingen, Dresden, Münster and San Diego. Redesign of the lab course in Theoretical Chemistry. |
| Jan 2007 – Dec 2010 | <b>Student Assistant, <i>Philipps-Universität Marburg</i></b><br>Supervision of lab courses in Physical Chemistry and Chemistry for Medical Students.                                                                                                                                                       |

## Third-party funding

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| 2024 | <b>Sustainable TiFe Generation from Secondary Sources</b><br>BMBF (Germany)   Principal Investigator   1.5 M€   3 years                                                                   |
| 2024 | <b>HIDA Visiting Researcher Grant (with S. Faouri)</b><br>Helmholtz Association   Host   10 k€   2 months                                                                                 |
| 2023 | <b>HIDA Visiting Researcher Grant (with A. Rowberg)</b><br>Helmholtz Association   Host   15 k€   3 months                                                                                |
| 2022 | <b>Generating Hydrogen Storage Materials from New Zealand Resources</b><br>BMBF (Germany)   Coordinator   400 k€   3 years                                                                |
| 2021 | <b>Establishing a German-New Zealand Research Centre for Green Hydrogen</b><br>BMBF (Germany)   Coordinator   760 k€   5 years                                                            |
| 2021 | <b>HIDA Visiting Scientist Grant (with M. Selzer)</b><br>Helmholtz Association   Fellow   10 k€   3 months                                                                                |
| 2017 | <b>Humboldt Return Fellowship</b><br>Alexander von Humboldt Foundation   Fellow   47 k€   1 year                                                                                          |
| 2016 | <b>Computation of highly accurate interatomic potentials of heavy and superheavy elements</b><br>New Zealand eScience Infrastructure (NeSI)   Principal Investigator   117 k€   1.5 years |
| 2014 | <b>Hochpräzise quantenchemische Simulationen von atomaren und molekularen Phasen unter extremen Bedingungen</b><br>Alexander von Humboldt Foundation   Fellow   85 k€   2 years           |

## Conference organization

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| 2025 | <b>E-MRS Fall Meeting 2025 — Symposium A</b><br>Co-organizer, Warsaw, Poland                                                                             |
| 2024 | <b>E-MRS Fall Meeting 2024 — Symposium C: Sustainable materials for chemical and electrochemical energy storage II</b><br>Main organizer, Warsaw, Poland |
| 2023 | <b>Inaugural New Zealand Hydrogen Symposium</b><br>Co-organizer, Dunedin, New Zealand                                                                    |
| 2015 | <b>Bonding Analysis Workshop</b><br>Conceptualization and main organizer, Marburg, Germany                                                               |

## Teaching

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| 2025 SoSe | <b>Hydrogen Technology</b><br>Co-lecturer, Technische Universität Hamburg (TUHH), Hamburg |
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| 2023 – present | <b>Lectureship</b><br>Lecturer, Technische Hochschule Mittelhessen, Gießen                                                                              |
| 2019 – 2019    | <b>Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa (Volkswagen-Stiftung)</b><br>Tutor, University of Mauritius, Reduit     |
| 2014 – 2019    | <b>Hands-on workshops on electronic structure methods</b><br>Workshop lecturer, Various universities, Hyderabad, Dunedin, Denton, Mariapfarr, Mauritius |
| 2010 – 2014    | <b>Practical course in Theoretical Chemistry</b><br>Teaching assistant, Philipps-Universität Marburg, Marburg                                           |
| 2007 – 2010    | <b>Physical Chemistry and Chemistry for Medical Students</b><br>Student assistant, Philipps-Universität Marburg, Marburg                                |

## Talks and posters

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| May 2025 | <b>German-New Zealand WTZ Meeting 2025</b> , Wellington, New Zealand<br>Invited Talk                                                                 |
| Feb 2025 | <b>11th International Conference on Advanced Materials and Nanotechnology (AMN11)</b> , Christchurch, New Zealand<br>Invited Talk                    |
| Jul 2024 | <b>New Zealand-Germany Energy Dialogue 2024</b> , Berlin, Wellington, Germany, New Zealand<br>Invited Talk                                           |
| Feb 2023 | <b>10th International Conference on Advanced Materials and Nanotechnology (AMN10)</b> , Rotorua, New Zealand<br>Invited Talk                         |
| Jul 2018 | <b>18th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2018)</b> , Rota, Spain<br>Invited Talk |
| Jul 2016 | <b>16th International Conference on Computational and Mathematical Methods in Science and Engineering (CMMSE-2016)</b> , Rota, Spain<br>Invited Talk |
| Mar 2024 | <b>German Physical Society (DPG) Spring Meeting 2024</b> , Berlin, Germany<br>Contributed Talk                                                       |
| Jan 2024 | <b>2nd New Zealand Hydrogen Symposium (NZHS-2)</b> , Wellington, New Zealand<br>Contributed Talk                                                     |
| Mar 2023 | <b>German Physical Society (DPG) Spring Meeting 2023</b> , Dresden, Germany<br>Contributed Talk                                                      |
| Sep 2022 | <b>European Materials Research Society (E-MRS) Fall Meeting 2022</b> , Warsaw, Poland<br>Contributed Talk                                            |
| Mar 2019 | <b>Chemiedozententagung 2019</b> , Koblenz, Germany<br>Contributed Talk                                                                              |
| Nov 2017 | <b>17th Biennial Symposium of the Australian and New Zealand Associations of von Humboldt Fellows</b> , Wellington, New Zealand<br>Contributed Talk  |
| Sep 2017 | <b>12th International Conference on Relativistic Effects in Heavy-Element Chemistry and Physics (REHE)</b> , Marburg, Germany<br>Contributed Talk    |

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| Aug 2016 | <b>New Zealand Institute of Chemistry Conference (NZIC-16)</b> ,<br>Queenstown, New Zealand<br>Contributed Talk                |
| May 2024 | <b>18th International Symposium of Metal-Hydrogen Systems (MH 2024)</b> , Saint-Malo, France<br>Poster                         |
| Sep 2018 | <b>54th Symposium on Theoretical Chemistry (STC)</b> , Halle (Saale),<br>Germany<br>Poster                                     |
| Aug 2017 | <b>11th World Association of Theoretical and Computational Chemists (WATOC)</b> , Munich, Germany<br>Poster                    |
| Jul 2016 | <b>9th Congress of the International Society for Theoretical Chemical Physics (ISTCP IX)</b> , Grand Forks (ND), USA<br>Poster |
| Sep 2015 | <b>51st Symposium on Theoretical Chemistry (STC)</b> , Postdam, Germany<br>Poster                                              |
| Sep 2013 | <b>49th Symposium on Theoretical Chemistry (STC)</b> , Erlangen, Germany<br>Poster                                             |
| Sep 2012 | <b>48th Symposium on Theoretical Chemistry (STC)</b> , Karlsruhe, Germany<br>Poster                                            |
| Oct 2011 | <b>Modeling of Molecular Properties Conference (MolMod)</b> , Heidelberg, Germany<br>Poster                                    |
| Jul 2011 | <b>9th World Association of Theoretical and Computational Chemists (WATOC)</b> , Santiago de Compostela, Spain<br>Poster       |
| Sep 2010 | <b>46th Symposium on Theoretical Chemistry (STC)</b> , Münster, Germany<br>Poster                                              |
| Feb 2025 | <b>University of Waikato</b> , Hamilton, New Zealand<br>Invited Seminar                                                        |
| Jun 2024 | <b>University of Hamburg</b> , Hamburg, Germany<br>Invited Seminar                                                             |
| Feb 2024 | <b>Massey University</b> , Auckland, New Zealand<br>Invited Seminar                                                            |
| Feb 2024 | <b>University of Waikato</b> , Hamilton, New Zealand<br>Invited Seminar                                                        |
| Mar 2023 | <b>Massey University</b> , Auckland, New Zealand<br>Invited Seminar                                                            |
| Mar 2023 | <b>University of Auckland</b> , Auckland, New Zealand<br>Invited Seminar                                                       |
| Dec 2021 | <b>Karlsruhe Institute of Technology (KIT)</b> , Karlsruhe, Germany<br>Invited Seminar                                         |
| May 2019 | <b>Helmholtz-Zentrum für Material- und Küstenforschung</b> ,<br>Geesthacht, Germany<br>Invited Seminar                         |
| Apr 2019 | <b>Universität Graz</b> , Graz, Austria<br>Invited Seminar                                                                     |
| Feb 2019 | <b>Technische Universität Berlin</b> , Berlin, Germany<br>Invited Seminar                                                      |

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| Jan 2019 | <b>GSI Helmholtzzentrum für Schwerionenforschung</b> , Darmstadt, Germany<br>Invited Seminar                                                                                    |
| Feb 2018 | <b>University of Otago</b> , Dunedin, New Zealand<br>Invited Seminar                                                                                                            |
| Jun 2017 | <b>University of Otago</b> , Dunedin, New Zealand<br>Invited Seminar                                                                                                            |
| Jul 2016 | <b>University of North Texas</b> , Denton (TX), USA<br>Invited Seminar                                                                                                          |
| Oct 2019 | <b>Summer School/Workshop on Computational Chemistry “Knowledge for Tomorrow – Cooperative Research Projects in Sub-Saharan Africa”</b> , Reduit, Mauritius<br>Workshop Lecture |
| Mar 2019 | <b>XIIth Workshop on Modern Methods in Quantum Chemistry</b> , Mariapfarr, Austria<br>Workshop Lecture                                                                          |
| Dec 2017 | <b>University of Otago</b> , Dunedin, New Zealand<br>Workshop Lecture                                                                                                           |
| Jul 2016 | <b>University of North Texas</b> , Denton (TX), USA<br>Workshop Lecture                                                                                                         |
| Jan 2014 | <b>MCBR Winter School</b> , Hyderabad, India<br>Workshop Lecture                                                                                                                |

## Peer-reviewed journal articles

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70. “Evaluation of the synergetic effects of Ni and Cr in  $\text{TiFe}_{(1-x-y)}\text{Cr}_x\text{Ni}_y$  for enhanced hydriding thermodynamics”: E. Pericoli, V. Ferretti, E. Alvares, P. Jerabek, C. Pistidda, L. Pasquini, *Adv. Energy Sustain. Res.*, **7**, e70243, 2026.
69. “Relativistic effects are crucial for uraniumphosphorus bonding - a DFT study of cyclopentadienyluranium-phosphine complexes”: J.-E. Ahrens, P. Jerabek, L. Vondung, *Mol. Phys.*, e2611400, 2026.
68. “Role of Elemental Substitution on Hydrogen Incorporation in Characteristic Ti-Fe-Based Oxides”: A. J. E. Rowberg, S. Kang, K. Sellschopp, P. Jerabek, T. W. Heo, B. C. Wood, *ACS Appl. Energy Mater.*, **9**, 1242–1253, 2026.
67. “Leveraging atomic disorder to modulate hydrogen storage thermodynamics in compositionally complex intermetallics”: Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Interdiscip. Mater.*, **0**, 1-13, 2026.
66. “Predicting hydrogen storage capacity loss of Si-containing FeTi alloys through ab initio thermodynamic modeling”: L. Dinachandran, E. Alvares, K. Sellschopp, T. Klassen, P. Jerabek, A. L. Garden, *Acta Mater.*, **301**, 121514, 2025.
65. “TiAlNb alloy interatomic potentials - comparing passive and active machine learning techniques with MTP and DeepMD”: A. Chandran, A. Santhosh, P. Jerabek, R. C. Aydin, C. J. Cyron, *Front. Mater.*, **12**, 2025.
64. “Low-temperature regeneration of metal hydride after impurity gas exposure”: Y. Shang, A. Santhosh, M. Ante, L. Wimbert, P. Jerabek, T. Schupp, T. Klassen, C. Pistidda, *Acta Mater.*, **298**, 121398, 2025.
63. “High Ionic Conduction in Rb- and Cs-Mixed Cation Amide for Energy Storage”: T. T. Le, K. Sellschopp, F. Murgia, A. L. Garden, S. Bordignon, J. P. Embs, M. R. Chierotti, A. Schökel, F. Karimi, P. Jerabek, T. Klassen, C. Pistidda, *Small*, 2502943, 2025.

62. "A Thermodynamic Approach to Model Multicomponent FeTi-based Alloys for Hydrogen Storage": E. Alvares, A. J. E. Rowberg, K. Sellschopp, B. C. Wood, T. Klassen, P. Jerabek, C. Pistidda, *Scr. Mater.*, **259**, 116516, 2025.
61. "Multiscale Modeling the FeTi Metal-Hydride Interphase Boundaries – Quantification of Decoupled Chemo-Mechanical Energies": E. Alvares, K. Sellschopp, B. Wang, S. Kang, T. Klassen, T. W. Heo, T. Klassen, P. Jerabek, C. Pistidda, *npj Comput. Mater.*, **10**, 249, 2024.
60. "Comparative analysis of ternary TiAlNb interatomic potentials - moment tensor vs. deep learning approaches": A. Chandran, A. Santhosh, C. Pistidda, P. Jerabek, R. Aydin, C. J. Cyron, *Front. Mater.*, **11**, 1466793, 2024.
59. "Thermodynamic assessment of the Ce-H and CeNi<sub>5</sub>-H system": P. Hannappel, E. Alvares, F. Heubner, C. Pistidda, P. Jerabek, T. Weißgärber, *Calphad*, **85**, 102701, 2024.
58. "Predictable Electronic Tuning of FeII versus RuII Complexes via Choice of Azine Donor - An Electrochemical and Theoretical Study": M. Robb, L. Bondi, S. Rodriguez-Jimenez, A. Garden, P. Jerabek, S. Brooker, *Dalton Trans.*, **53**, 1999-2007, 2024.
57. "Tipping the balance between the bcc and fcc phase within the alkali and coinage metal groups": A. Robles-Navarro, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **63**, e202313679, 2024.
56. "First-principles study on interfacial property in MgB<sub>2</sub>-based reactive hydride composites": Y. Shang, A. Santhosh, P. Jerabek, T. Klassen, C. Pistidda, *Scr. Mater.*, **240**, 115837, 2024.
55. "Low-Cost, Multifunctional, and Sustainable Sodium Sulfido Ferrate(II)": M. Ghazanfari, K. Siemensmeyer, A. Santhosh, J. Vrijmoed, M. Tallu, S. Dehnen, P. Jerabek, G. Thiele, *Inorg. Chem.*, **62**, 15358-15366, 2023.
54. "Ultra-lightweight compositionally complex alloys with large ambient-temperature hydrogen storage capacity": Y. Shang, Z. Lei, E. Alvares, S. Garroni, R. Dore, M. Rustici, S. Enzo, A. Schökel, Y. Shi, P. Jerabek, Z. Lu, T. Klassen, C. Pistidda, *Mater. Today*, **67**, 113-126, 2023.
53. "Influence of near-surface oxide layers on TiFe hydrogenation - Mechanistic insights and implications for hydrogen storage applications": A. Santhosh, S. Kang, N. Keilbart, B. C. Wood, T. Klassen, P. Jerabek, M. Dornheim, *J. Mater. Chem. A*, **11**, 18776-18789, 2023.
52. "Experimental and Computational Studies on the Formation of Mixed Amide-Hydride Solid Solutions for CsNH<sub>2</sub>-CsH System": T.-T. Le, A. Santhosh, S. Bordignon, M. R. Chierotti, P. Jerabek, T. Klassen, C. Pistidda, *Results in Engineering*, **17**, 100895, 2023.
51. "Magnesium- and intermetallic alloys-based hydrides for energy storage: modelling, synthesis and properties": L. Pasquini, K. Sakaki, E. Akiba, P. Jerabek et al., *Prog. Energy*, **4**, 32007, 2022.
50. "Relativistic Effects Stabilize Unusual Gold(II) Sulfate Structure via Auophilic Interactions": P. Jerabek, A. Santhosh, P. Schwerdtfeger, *Inorg. Chem.*, **61**, 13077-13084, 2022.
49. "Solving a problem with a single parameter: A smooth bcc to fcc phase transition for metallic lithium": P. Jerabek, A. Burrows, P. Schwerdtfeger, *Chem. Commun.*, **58**, 13369-13372, 2022.
48. "Remarkable Infrared falsenlinear Optical, Dielectric, and Strong Diamagnetic Characteristics of Semiconducting K<sub>3</sub>[BiS<sub>3</sub>]": M. R. Ghazanfari, L. Vittadello, D. Al-Sabbagh, A. Santhosh, C. Frankcom, F. Fuß, C. A. von Randow, K. Siemensmeyer, J. C. Vrijmoed, F. Emmerling, P. Jerabek, M. Imlau, G. Thiele, *J. Phys. Chem. Lett.*, **13**, 6987-6993, 2022.

47. "Large-scale synthesis of mixed valance K<sub>3</sub>[Fe<sub>2</sub>S<sub>4</sub>] with high dielectric and ferrimagnetic characteristics": M. R. Ghazanfari, A. Santhosh, J. C. Vrijmoed, K. Siemensmeyer, B. Peters, S. Dehnen, P. Jerabek, G. Thiele, *RSC Adv.*, **12**, 30514–30521, 2022.
46. "Large Exchange Bias, High Dielectric Constant, and Outstanding Ionic Conductivity in a Single-Phase Spin Glass": M. R. Ghazanfari, A. Santhosh, K. Siemensmeyer, F. Fuß, L. Staab, J. C. Vrijmoed, B. Peters, M. Liesegang, S. Dehnen, O. Oeckler, P. Jerabek, G. Thiele, *Adv. Electron. Mater.*, **8**, 2200483, 2022.
45. "From the gas phase to the solid state: The chemical bonding in the superheavy element flerovium": E. Florez, O. R. Smits, J.-M. Mewes, P. Jerabek, P. Schwerdtfeger, *J. Chem. Phys.*, **157**, 64304, 2022.
44. "An Effective Activation Method for Industrially Produced TiFeMn Powder for Hydrogen Storage": D. M. Dreistadt, L. Thi-Thu, G. Capurso, J. M. Bellosta von Colbe, A. Santhosh, C. Pistidda, N. Scharnagl, H. Ovri, C. Milanese, P. Jerabek, T. Klassen, J. Jepsen, *J. Alloys Compd.*, **919**, 165847, 2022.
43. "Quantitative assessment of ligand substituent effects on  $\sigma$ - and  $\pi$ -contributions to Fe-N bonds in spin crossover FeII complexes": L. Bondi, A. L. Garden, F. Totti, P. Jerabek, S. Brooker, *Chem. Eur. J.*, **28**, e202104314, 2022.
42. "Modeling the thermodynamics of the FeTi hydrogenation under para-equilibrium: An ab-initio and experimental study": E. Alvares, P. Jerabek, Y. Shang, A. Santhosh, C. Pistidda, T. W. Heo, B. Sundman, M. Dornheim, *Calphad*, **77**, 102426, 2022.
41. "Hydrogenation via a low energy mechafalsechemical approach: The MgB<sub>2</sub> case": C. Pistidda, A. Santhosh, P. Jerabek, Y. Shang, A. Girella, C. Milanese, M. Dore, S. Garroni, S. Bordigfalsen, M. R. Chierotti, T. Klassen, M. Dornheim, *J. Phys. Energy*, **3**, 44001, 2021.
40. "Oganesson: A Noble Gas Element That Is Neither Noble Nor a Gas": O. R. Smits, J. Mewes, P. Jerabek, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **59**, 23636–23640, 2020.
39. "First-principles melting of krypton and xefalsen based on many-body relativistic coupled-cluster interaction potentials": O. R. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Phys. Rev. B*, **101**, 104103, 2020.
38. "Catalytic Asymmetric Fluorination of Copper Carbene Complexes: Preparative Advances and a Mechanistic Rationale": M. Buchsteiner, L. Martinez-Rodriguez, P. Jerabek, I. Pozo, M. Patzer, N. Nöthling, C. W. Lehmann, A. Fürstner, *Chem. Eur. J.*, **26**, 2509–2515, 2020.
37. "Quantitative and Chemically Intuitive Evaluation of the Nature of M–L Bonds in Paramagnetic Compounds: Application of EDA-NOCV Theory to Spin Crossover Complexes": L. Bondi, A. L. Garden, P. Jerabek, F. Totti, S. Brooker, *Chem. Eur. J.*, **26**, 13677–13685, 2020.
36. "Ligands Based on Phosphine-Stabilized Aluminum(I), Boron(I), and Carbon(0)": L. Vondung, P. Jerabek, R. Langer, *Chem. Eur. J.*, **25**, 3068–3076, 2019.
35. "Oganesson Is a Semiconductor: On the Relativistic Band-Gap Narrowing in the Heaviest falseble-Gas Solids": J.-M. Mewes, P. Jerabek, O. R. Smits, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **58**, 14260–14264, 2019.
34. "Solid Oganesson via a Many-Body Interaction Expansion Based on Relativistic Coupled-Cluster Theory and from Plane-Wave Relativistic Density Functional Theory": P. Jerabek, O. R. Smits, J.-M. Mewes, K. A. Peterson, P. Schwerdtfeger, *J. Phys. Chem. A*, **123**, 4201–4211, 2019.
33. "Dative and Electron-Sharing Bonding in Transition Metal Compounds": P. Jerabek, P. Schwerdtfeger, G. Frenking, *J. Comput. Chem.*, **40**, 247–264, 2019.

32. "A Hundred-Year-Old Experiment Re-evaluated: Accurate Ab-Initio Monte-Carlo Simulations of the Melting of Radon": O. Smits, P. Jerabek, E. Pahl, P. Schwerdtfeger, *Angew. Chem. Int. Ed.*, **57**, 9961–9964, 2018.
31. "The Light-Driven Isomerization of Aqueous Nitrate: A Theoretical Perspective": J.-M. Mewes, P. Jerabek, D. S. Bohle, P. Schwerdtfeger, *ChemPhotoChem*, **2**, 725–733, 2018.
30. "Tipping the Balance between Ligand and Metal Protonation due to Relativistic Effects: Unusually High Proton Affinity in Gold(I) Pincer Complexes": P. Jerabek, L. Vondung, P. Schwerdtfeger, *Chem. Eur. J.*, **24**, 6047–6051, 2018.
29. "A relativistic coupled-cluster interaction potential and rovibrational constants for the xenon dimer": P. Jerabek, O. Smits, E. Pahl, P. Schwerdtfeger, *Mol. Phys.*, **116**, 1–8, 2018.
28. "Static dipole polarizability of palladium from relativistic coupled-cluster theory": P. Jerabek, P. Schwerdtfeger, J. K. Nagle, *Phys. Rev. A*, **98**, 12508, 2018.
27. "Electron and nucleon localization functions of oganesson: Approaching the Thomas-Fermi limit": P. Jerabek, B. Schuettrumpf, P. Schwerdtfeger, W. Nazarewicz, *Phys. Rev. Lett.*, **120**, 53001, 2018.
26. "Suppressed Phosphine Dissociation by Polarization Effects on the Donor–Acceptor Bonds in  $[\text{Ni}(\text{PEt}_3)_{4-n}(\text{ECp}^*)_n]$  ( $\text{E} = \text{Al}, \text{Ga}$ )": J. Hornung, J. Weßing, P. Jerabek, C. Gemel, A. Pöthig, G. Frenking, R. A. Fischer, *Inorg. Chem.*, **57**, 12657–12664, 2018.
25. "Influence of Relativistic Effects on Bonding Modes in M(II) Dinuclear Complexes ( $\text{M} = \text{Au}, \text{Ag}$ , and  $\text{Cu}$ )": P. Jerabek, B. von der Esch, H. Schmidbaur, P. Schwerdtfeger, *Inorg. Chem.*, **56**, 14624–14631, 2017.
24. "The Oxygen-Rich Beryllium Oxides  $\text{BeO}_4$  and  $\text{BeO}_6$ ": Q. Zhang, P. Jerabek, M. Chen, M. Zhou, G. Frenking, *Angew. Chem. Int. Ed.*, **55**, 10863–10867, 2016.
23. "Understanding the F 1s NEXAFS Dichroism in Fluorinated Organic Semiconductors": M. Klues, P. Jerabek, T. Breuer, M. Oehzelt, K. Hermann, R. Berger, G. Witte, *J. Phys. Chem. C*, **120**, 12693–12705, 2016.
22. "Zn...Zn interactions at nickel and palladium centers": K. Freitag, M. Molon, P. Jerabek, K. Dilchert, C. Rösler, R. W. Seidel, C. Gemel, G. Frenking, R. A. Fischer, *Chem. Sci.*, **7**, 6413–6421, 2016.
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